The BIND Lightcone Suite I: Baryon-Painted Weak-Lensing and Sunyaev–Zel'dovich Lightcones from Dark-Matter-Only Simulations

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Abstract

Baryonic feedback is now the dominant systematic limiting small-scale weak-lensing (WL) and thermal Sunyaev–Zel'dovich (tSZ) cosmology, and existing correction schemes force a choice between physical fidelity (full hydrodynamical simulations, computationally prohibitive at survey volume) and speed (analytic baryon correction models, which impose spherically symmetric, deterministic halo profiles). We present BIND, a conditional flow-matching emulator trained on the CAMELS-TNG suite that paints stochastic, multi-channel baryonic fields (gas and stellar density, Compton-y, temperature, entropy, electron pressure) onto individual dark-matter-only (DMO) halos, and describe its extension to a full cosmological lightcone: per-snapshot painting of TNG300-Dark halos, circular-aperture compositing, lens-plane construction, and multi-plane ray-tracing that produces convergence $\kappa(z_{\rm s})$, Compton-y, and kinetic-SZ optical depth τ maps self-consistently from a single painted density field. At fiducial CAMELS parameters, the BIND lightcone reproduces the true TNG300 hydrodynamical lightcone's WL power spectrum and Minkowski functionals to a few percent ($C_\ell^{\kappa\kappa,\rm BIND}/C_\ell^{\kappa\kappa,\rm hydro} \approx 1.00$; $R(\nu)_{\rm BIND}/R(\nu)_{\rm hydro} = 0.98$–$1.06$), with peak and minimum counts agreeing within their (larger, small-number-dominated) sampling scatter in the sparsely populated tails, and halo-level Y–M and pressure-profile statistics agreeing within overlapping intrinsic-scatter bands. We show that the peak-stacked ratio $R(\nu) \equiv \langle y\rangle/\langle\kappa\rangle$ is not a pure gas fraction but a specific-thermal-energy proxy, $R(\nu) \propto f_{\rm gas}\, T_{\rm mw}$ (Pearson $r = 0.79$), and we establish a direct, quantitative bridge between per-halo gas astrophysics and field-level WL observables: the background-subtracted group-scale gas fraction predicts the lightcone power-spectrum suppression $S(\ell)$ with $r = 0.91$, and the integrated Compton-Y_{500c} is an even tighter predictor ($r = 0.93$). We characterize the completeness of the reuse-based low-mass (10^{12}–$10^{13}\,{\rm M}_\odot/h$) extension and lay out the mass-resolution validation program still required before extrapolating beyond the TNG300-Dark training resolution. This pipeline, its released weights, and the validated map products form the methodological foundation for the feedback-response, data-application, kinetic-SZ, and field-level-anisotropy studies presented in the companion papers of this series.

1 Introduction

Weak gravitational lensing (WL) and the thermal Sunyaev–Zel'dovich (tSZ) effect are two of the most direct cosmological probes of the matter and thermal-energy distribution in the late-time Universe, and both are increasingly limited not by statistical power but by our ignorance of baryonic feedback. Active galactic nucleus (AGN) and stellar-driven winds evacuate gas from galaxy-group and cluster halos, redistributing mass on scales up to several virial radii and suppressing the non-linear matter power spectrum by tens of percent at $k \gtrsim 1\,h\,{\rm Mpc}^{-1}$ relative to a dark-matter-only

(DMO) universe (van Daalen et al., 2011, 2020). Recent multi-wavelength observations sharpen the concern that current-generation hydrodynamical simulations under-predict the strength of this feedback: joint kinetic-SZ (kSZ) and X-ray analyses find that real group-scale halos expel gas more efficiently, and to larger radii, than simulations such as FLAMINGO predict (Siegel et al., 2025), and the cross-correlation of DES Year-3 shear with ACT tSZ maps is detected at high significance with evidence for reduced thermal pressure in halos below $M \sim 10^{14}\,\mathrm{M}_{\odot}\,h^{-1}$ (Pandey et al., 2025; Bigwood et al., 2024). Analytic baryon correction models (BCMs), long used to marginalize this systematic for two-point cosmic-shear analyses at the ∼percent level (Schneider and Teyssier, 2015; Chisari et al., 2019; Mead et al., 2021), are only as good as the parametric, spherically symmetric halo profiles they assume.

The problem sharpens further once Stage-IV surveys (LSST, *Euclid*, the *Roman Space Telescope*) move beyond the power spectrum to exploit the non-Gaussian information encoded in the WL convergence field — peak and minimum counts, Minkowski functionals, and the one-point PDF, which can improve cosmological constraining power by factors of several (Martinet et al., 2021; Gatti et al., 2022; Zürcher et al., 2023). Baryonic corrections to these statistics are not simple rescalings of the power-spectrum suppression: peaks are dominated by the cores of massive halos, where AGN feedback creates pressure cavities, while the power spectrum and minima are controlled by group-scale halos and large-scale gas redistribution. A systematic study of BCM-corrected peak counts against matched hydrodynamical simulations found that BCMs make two compensating errors — underpredicting core masses while overpredicting large-radius suppression — that bias cosmological inference even when the power spectrum is fit correctly, and that this failure sets in already by moderate peak significance (Lee et al., 2022). No existing framework simultaneously captures this halo-scale stochasticity and produces self-consistent multi-probe (WL + tSZ) lightcones.

Field-level baryonification and machine-learned emulators close part of this gap. Neural-network field-level inference trained on CAMELS hydrodynamical simulations can robustly marginalize over baryonic uncertainty when extracting cosmological parameters from mass maps (Villaescusa-Navarro et al., 2021), and baryonification-calibrated emulators such as BACCO reach percent-level accuracy on the matter power spectrum, though calibrated on a dedicated baryonification-algorithm suite rather than trained on CAMELS (Aricò et al., 2021); map-level baryonification has also been extended to jointly paint WL and tSZ fields on the full sky within a BCM framework (Anbajagane et al., 2024). These approaches, however, either remain tied to a 2-point or halo-profile representation of the correction, or do not natively produce a stochastic, per-halo realization with the full thermodynamic channel set (density, temperature, entropy, pressure) needed to jointly predict κ, y, and kSZ observables from one consistent painted field. BIND closes this gap: a conditional flow-matching model, trained end-to-end on CAMELS-TNG halos with a 35-dimensional subgrid-physics parameterization (SB35; Villaescusa-Navarro et al., 2023), that paints stochastic multi-channel baryonic fields onto individual DMO halos and can therefore be applied, halo-by-halo, to an arbitrary DMO lightcone at negligible marginal cost per feedback variation (Lee, 2026a).

This paper is the first in a five-part series and establishes the pipeline that the rest of the series builds on. It is deliberately narrow in scope: we describe the BIND lightcone construction end-to-end (Section 2), validate the resulting κ, Compton-y, and τ maps against the true TNG300 hydrodynamical lightcone at fixed (fiducial) subgrid parameters (Section 3, Results 1), diagnose what the tSZ-to-WL peak ratio $R(\nu)$ physically measures (Results 2), and establish the quantitative bridge linking per-halo gas content to the field-level WL suppression (Results 3), which is the load-bearing result the companion papers use to translate observational gas-fraction constraints into WL systematics and vice versa. We close with a characterization of the pipeline's completeness at group and sub-group halo mass (Results 4) and an honest accounting of its current limitations

(Section 4). We do not attempt, in this paper, the systematic feedback-parameter study, the survey-data application, the velocity-resolved kSZ extension, or the field-level anisotropy analysis that the companion papers (Lee, 2026b,c,d,e, Papers II–V) build on this foundation.

2 Methods

2.1 The bind emulator

BIND is a conditional flow-matching generative model that maps a DMO density cutout around an individual halo, together with a 35-dimensional vector of cosmological and subgrid-physics parameters drawn from the CAMELS-TNG SB35 design (Villaescusa-Navarro et al., 2023), to the corresponding hydrodynamical fields at that halo: dark-matter response, gas density, and stellar density, plus (in the configuration used throughout this paper) four gas-thermodynamic channels — Compton-y, temperature, entropy, and electron pressure — trained jointly with the mass channels.[1] Conditional flow matching (Lipman et al., 2023) learns a continuous, noise-to-data velocity field rather than a fixed deterministic mapping, so that repeated draws from the same conditioning (DMO halo + parameters) sample the physical halo-to-halo scatter present in the training simulations, rather than returning a single mean profile as a BCM would. Because the conditioning vector spans the full SB35 subgrid design, the emulator can be evaluated at any point in that 35-dimensional feedback space without re-running a hydrodynamical simulation, which is what makes a systematic, per-parameter lightcone study (Lee, 2026b, Paper II) computationally feasible in the first place. BIND operates on 2D projected 128×128 halo cutouts; the lightcone extension described below applies this halo-by-halo painting to full-box DMO snapshots and assembles the result into a ray-traced sky.

2.2 Lightcone construction

The lightcone pipeline (`src/bind/inference/{science,pipeline,lensplane, truth_lightcone}.py`) builds a BIND-painted, multi-plane-ray-traced sky from the TNG300-DARK DMO simulation (box size $205\,h^{-1}$Mpc) in six stages.

1. **Per-snapshot projection (Stage 1).** Twenty TNG300-Dark snapshots, chosen to tile the observer's past lightcone from low to high redshift, are each projected into comoving z-slabs, and every friends-of-friends (FoF) halo above the painting mass floor is cut out as a DMO patch. Twenty snapshots are used,[2] and the resulting twenty comoving shells span $z = 0.034$–2.444.

2. **Halo-by-halo painting (Stage 2).** BIND generates the multi-channel hydro patch for each halo cutout conditioned on the fiducial (or varied) 35-parameter vector, producing a stochastic per-halo realization of gas, stars, DM response, and gas-thermodynamic fields.

3. **Compositing.** Painted patches are pasted back into the full-box projected slab with a circular, Hann-tapered aperture of radius $4 \times R_{200c}$, the pipeline default (`r200_factor=4.0`), rather than the legacy square taper, which smears flux across halo boundaries.

[1] The model architecture, training procedure, and per-halo validation against held-out CAMELS hydrodynamical realizations are described in the companion model paper (Lee, 2026a); we summarize only what is needed to understand the lightcone pipeline.

[2] Snapshot indices 96, 90, 85, 80, 76, 71, 67, 63, 59, 56, 52, 49, 46, 43, 41, 38, 35, 33, 31, 29, ordered low-z first.

4. **Lens planes.** The composited mass slabs are converted to lens planes via a 2D Poisson solve in Fourier space that exactly reproduces the convention used by the external ray-tracer `lux`, so that lens planes written by this pipeline are drop-in replacements for planes `lux` would generate from a native hydrodynamical run.

5. **Multi-plane ray tracing.** `lux` ray-traces the stack of lens planes to produce the convergence $\kappa(z_s)$ at each of five tomographic source planes. The tSZ Compton-y field is *also* sampled at the deflected ray position on each plane (not simply summed along the undeflected line of sight), so the tSZ lightcone is genuinely ray-traced, not a Born-approximation slab sum.

6. **Truth lightcone and paired DMO trace.** For validation, the identical pipeline is run on the true IllustrisTNG hydrodynamical fields (DM response, gas, stars, and the four thermo channels) at the *same* DMO-FoF halo positions and under the *same* lightcone geometry, producing a `truth` composite that is a drop-in replacement for the BIND composite in the downstream recomposite → lens-plane → ray-trace → statistics chain. A paired DMO-only trace (no baryonic painting) is built from the same Stage-1 maps for use as the common-mode reference in suppression ratios $S(\ell)$ (Section 2.3).

The field of view is $5 \times 5\,\mathrm{deg}^2$ by default. Our fiducial-closure test (Results 3.1) uses 50 independent lightcone realizations at identical geometry to characterize the sample-variance envelope.

Optical depth / electron column. The kSZ optical-depth (equivalently, the fast-radio-burst dispersion measure, up to a constant) plane is built by the same ray-tracing machinery as the y-plane, additively accumulating $\tau = \sigma_T\, x_e\, \Sigma_{\rm gas}/m_p$ along the deflected ray, with no lensing kernel (optical depth is not a lensing observable). This is a genuinely velocity-free electron-column estimate: it does not include the kinematic (velocity-weighted) kSZ temperature decrement, which requires a gas-velocity field BIND does not currently paint (Section 4). The choice is deliberate and matches the way the kSZ observable is actually extracted from survey data: the velocity dependence factorizes out of the standard pairwise/stacked kSZ estimator, so that the quantity a lightcone pipeline needs to predict at fixed halo mass is precisely the optical-depth (gas-column) profile (Schaan et al., 2021; Hadzhiyska et al., 2024; Ried Guachalla et al., 2025).

2.3 Map suite

The pipeline produces, from a single painted lightcone, three co-registered observable planes: (i) tomographic convergence $\kappa(z_s)$ at five source redshifts, $z_s = 0.5, 1.0, 1.5, 2.0, 2.44$; (ii) the Compton-$y$ (tSZ) map; and (iii) the electron-column / kSZ optical depth τ map, equivalently a fast-radio-burst dispersion measure up to the constant DM = τ/TAU_PER_DM (Section 2.2). Figure 1 shows the BIND $\kappa(z_s = 1)$, the paired DMO-only κ, and the BIND Compton-y map from the same fiducial realization: the baryonified field is visually distinct from its DMO progenitor, most apparent as an overall smoothing of the diffuse small-scale texture across the field (the DMO panel is visibly grainier) rather than a localized change at any individual peak, consistent with feedback redistributing mass broadly rather than only at halo centers; the y map traces halo cores with the expected compact morphology.

2.4 Statistics pipeline

Field-level statistics are computed with a common module (`src/bind/inference/stats.py`) applied identically to BIND, truth, and DMO lightcones so that all comparisons are same-pipeline ratios. The statistic set comprises: the tomographic angular power spectrum C_ℓ (auto and cross,

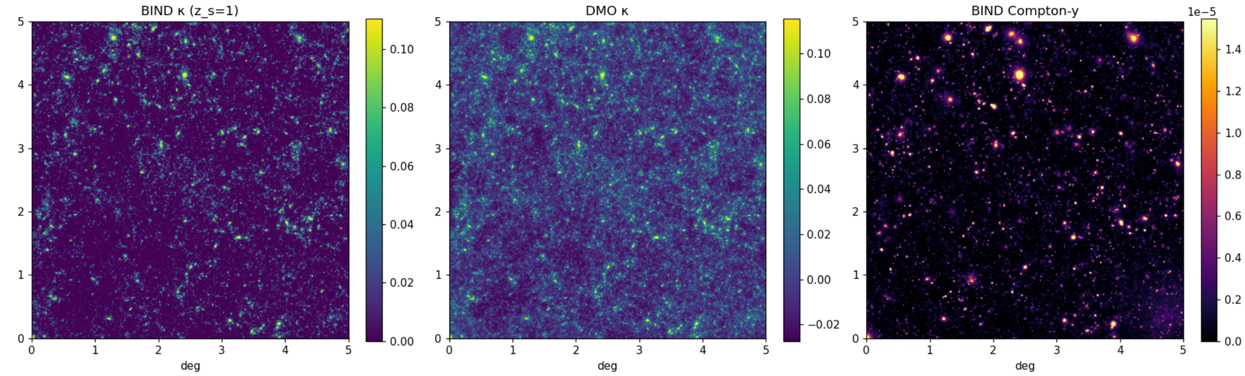

Figure 1: Fiducial BIND lightcone maps over the full $5 \times 5\,\mathrm{deg}^2$ field of view. *Left:* BIND convergence κ at source redshift $z_s = 1.0$. *Middle:* the paired DMO-only convergence at identical geometry, used as the common-mode reference for all suppression ratios in this paper. *Right:* the BIND Compton-y map from the same realization. The painted field is visually distinct from the DMO progenitor as an overall smoothing of the field-wide small-scale texture (most apparent in the diffuse, non-halo regions) rather than a localized change at individual peaks, and the y map traces halo cores with compact, feedback-sculpted morphology — the qualitative signature the quantitative validation in Figure 2 confirms statistically.

including $\kappa \times y$ and the tSZ auto-spectrum C_ℓ^{yy}) and the derived suppression $S(\ell) = C_\ell^{\mathrm{BIND}}/C_\ell^{\mathrm{DMO}}$; peak and minimum counts as a function of signal-to-noise threshold ν (8-neighbour local-extremum definition, with optional Gaussian shape noise added before smoothing); non-Gaussian statistics (the one-point PDF, variance/skewness/kurtosis, and the three Minkowski functionals V_0 (area), V_1 (perimeter), V_2 (genus) as functions of threshold); the peak-stacked ratio $R(\nu) = \langle y \rangle / \langle \kappa \rangle$ at WL peaks (Section 3.2); and per-halo scaling relations (Y_{500c}, f_{gas}, $f_\star$, mass-weighted temperature T_{mw}). The same per-halo machinery additionally yields three mass-binned diagnostics used in Section 3.1 and Figure 3: the stacked electron-column ratio $\tau_{\mathrm{BIND}}/\tau_{\mathrm{hydro}}$, the radially-resolved projected pressure profile $y(r)/y(r_{500c})$ compared against the Arnaud et al. universal profile (Arnaud et al., 2010), and the per-halo scatter $\sigma(\log Y \mid M)$ at fixed halo mass. Angular power spectra are computed with a CIC-based mass-assignment estimator (Pylians `Pk_plane`/`XPk_plane`) using the field-of-view in radians as the box size, so that the returned Fourier wavenumber is directly the multipole ℓ. Signal-to-noise bins extend to $\nu = 12$ (extended from an earlier $\nu \leq 6$ cutoff that discarded the highest-significance peaks).

Raw CIC mass assignment introduces a well-known aliasing artifact that appears as a spurious upturn in $C_\ell^{\kappa\kappa}$ at high multipole; an optional interlacing + deconvolution correction (`mas_correct=True`) is unit-validated (on synthetic test data, not yet directly on the TNG300 lightcone maps) to remove it to 0.4% out to $0.8\,k_{\mathrm{Nyquist}}$, but is not applied by default because the artifact is common-mode between the BIND and DMO (or truth) pipelines and cancels in every ratio-based statistic used in this paper (Section 4).

3 Results

3.1 Fiducial validation: bind vs. TNG300 hydrodynamical truth

The central validation test compares the BIND-painted lightcone against the true TNG300 hydrodynamical lightcone, built by the identical pipeline from the same halos under the same lightcone

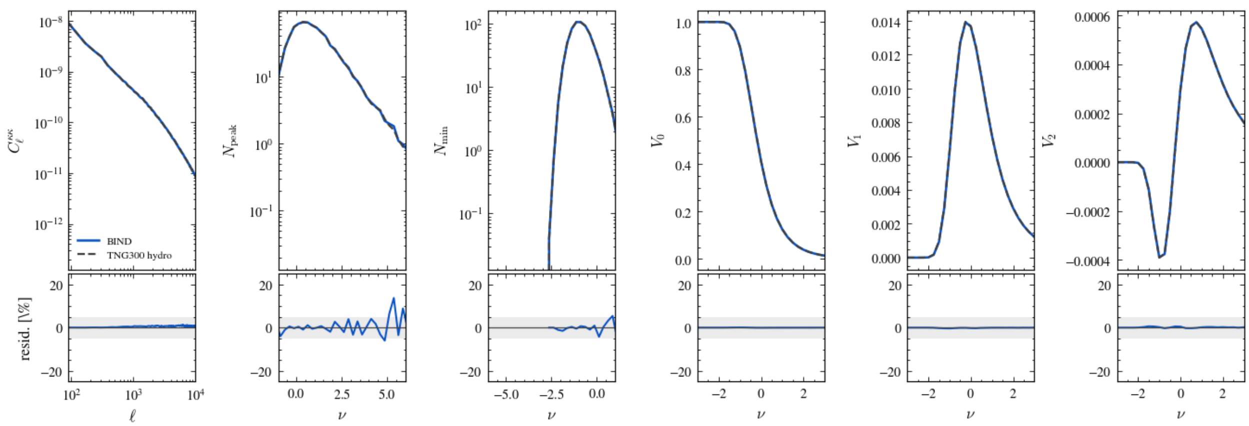

Figure 2: Field-level fiducial closure: BIND (solid) vs. TNG300 hydrodynamical truth (dashed) for the convergence power spectrum $C_\ell^{\kappa\kappa}$, peak counts $N_{\rm peak}(\nu)$, minimum counts $N_{\rm min}(\nu)$, and the three Minkowski functionals $V_0, V_1, V_2(\nu)$, with fractional residuals (shaded $\pm$ band) in the lower row. $C_\ell^{\kappa\kappa}$ and the Minkowski functionals agree to within a few percent across the full range shown; peak counts show larger ($\sim \pm 15\%$) but statistically consistent scatter in their sparsely populated high-ν tail, while minimum counts stay within $\sim \pm 7\%$ across their full range. This is the headline result of the paper: at fixed subgrid parameters, the BIND lightcone is statistically indistinguishable from the true hydrodynamical lightcone across the full non-Gaussian statistic set used by Stage-IV WL analyses.

geometry (Section 2.2), at fixed fiducial CAMELS-SB35 parameters. Figure 2 shows the field-level comparison across all six statistics in the pipeline (Section 2.4): the convergence power spectrum, peak counts, minimum counts, and the three Minkowski functionals, each with a residual sub-panel. The power-spectrum and Minkowski-functional residuals are flat to within a few percent across the full range plotted ($10^2 \lesssim \ell \lesssim 10^4$); peak counts are noisier, with residuals reaching $\sim \pm 15\%$ in the sparsely populated high-ν tail, while minimum-count residuals stay within $\sim \pm 7\%$ over their full plotted range; both are consistent with small-number statistics rather than a systematic mismatch. Averaged over 50 independent realizations at identical geometry, the convergence power ratio is $C_\ell^{\kappa\kappa,\rm BIND}/C_\ell^{\kappa\kappa,\rm hydro} \approx 1.00$, the baryonic suppression tracks the hydrodynamical suppression curve, the peak-stacked ratio $R(\nu)_{\rm BIND}/R(\nu)_{\rm hydro} = 0.98$–$1.06$ (rising by a factor of ~ 2 over the plotted ν range, in both BIND and truth), and the y-profile stacked at WL peaks matches to $\sim 5\%$.

Halo-level thermodynamic closure is shown in Figure 3. The Y–M relation tracks the TNG300 hydrodynamical relation across $M_{200c} = 10^{13}$– $10^{14.5}\ {\rm M}_\odot/h$ with overlapping intrinsic-scatter bands. The stacked electron-column ratio $\tau_{\rm BIND}/\tau_{\rm hydro}$ is ~ 1.05–1.15 for the two lowest M_{200c} bins ($10^{13.0}$–$10^{13.5}$ and $10^{13.5}$–$10^{14.0}$) and closer to unity for the highest bin, consistent with an independently computed group-scale offset of $+11\%$ reported from the electron-column validation script. Projected pressure profiles $y(r)/y(r_{500c})$ in four mass bins spanning $10^{13.0}$–$10^{14.2}\ {\rm M}_\odot/h$ match the Arnaud et al. universal pressure profile (Arnaud et al., 2010) to $\sim 2\,r_{500c}$, diverging at larger radius where the profile comparison is dominated by small-number and outer-radius noise.

The mass-condensation channel also reproduces per-halo gas-fraction and stochasticity statistics. BIND matches the truth $f_{\rm gas}(M_{200c})$ relation to 2–11%; at group scale ($M_{200c} > 5 \times 10^{13}\ {\rm M}_\odot/h$), $f_{\rm gas,500}^{\rm BIND} = 0.137$ vs. $f_{\rm gas,500}^{\rm truth} = 0.132$ ($\sim 4\%$ offset). The per-halo scatter in $\log Y$ at fixed mass is 0.23–0.44 dex in BIND vs. 0.24–0.41 dex in the hydrodynamical truth (a 3–6% excess at group scale, matched at cluster scale), while the scatter in $\log f_{\rm gas}$ at fixed mass is statistically identical be-

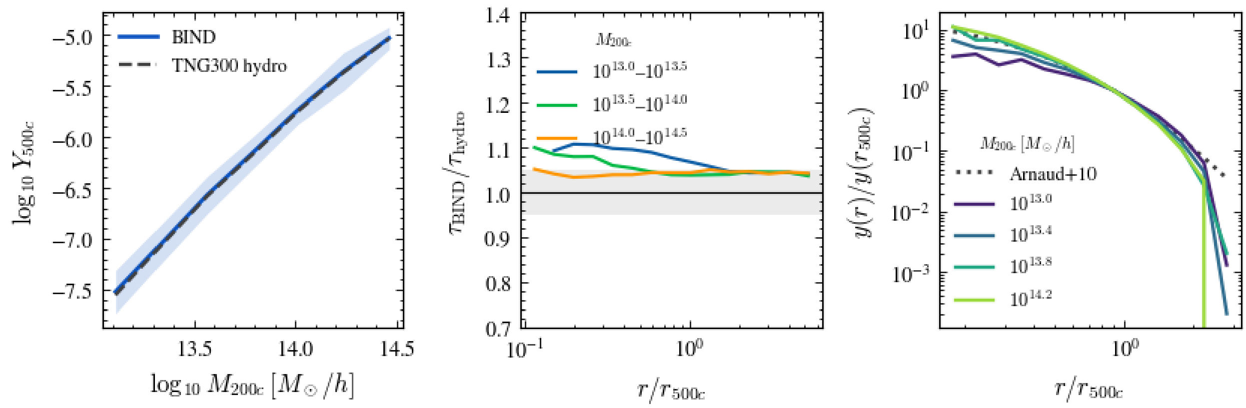

Figure 3: Halo-level thermodynamic closure at snapshot 96 ($z \approx 0.03$). *Left:* the Y_{500c}–M_{200c} relation, BIND (solid) vs. TNG300 hydrodynamical truth (dashed), with 1σ intrinsic-scatter bands overlapping across 10^{13}–$10^{14.5}\,\mathrm{M}_\odot/h$. *Middle:* the stacked electron-column ratio $\tau_{\rm BIND}/\tau_{\rm hydro}$ by mass bin, ~ 1.05–1.15 at group scale and closer to unity at the highest mass bin shown. *Right:* projected pressure profiles $y(r)/y(r_{500c})$ in four mass bins compared to the Arnaud et al. universal profile (Arnaud et al., 2010), matching to $\sim 2\,r_{500c}$.

tween BIND and truth. A separate, per-halo-atlas measurement at R_{500c} (rather than the map-level peak-stacking used above) finds a smaller scatter range, $\sigma(\log Y|M) = 0.12$–0.22 dex, consistent with a mildly over-dispersed generative model at group scale; we report both numbers with their distinct provenance (map-level peak-stacking vs. per-halo R_{500c} atlas) rather than merging them, since they are not measuring identical apertures. In either case the generative model reproduces — rather than suppresses — the halo-to-halo scatter that a deterministic BCM profile cannot capture by construction, which is the qualitative differentiator motivating a stochastic, per-halo emulator over a parametric correction.

Finally, Figure 4 shows the tomographic $\kappa \times y$ cross-spectrum and the tSZ auto-spectrum C_ℓ^{yy} at fiducial parameters across all five source redshifts, demonstrating that the ray-traced multi-probe map suite (Section 2.3) produces well-behaved cross-statistics suitable for joint WL $\times$ tSZ analysis (the subject of Lee (2026b, Paper II)). We revisit the known $\sim -12\%$ deficit in C_ℓ^{yy} relative to observed mean-y amplitudes in Section 4.

3.2 $R(\nu)$ tracks specific thermal energy, not gas fraction alone

The peak-stacked ratio $R(\nu) = \langle y\rangle/\langle\kappa\rangle$ was originally motivated as a projected gas-fraction proxy at WL peaks, since $y \propto$ pressure integrates gas mass and temperature while κ integrates total (mostly dark-matter) mass. The pipeline's own documentation makes the more precise statement explicit: since $Y \propto f_{\rm gas}\,M\,T_{\rm mw}$ at fixed halo mass, $R(\nu) \propto f_{\rm gas}\,T_{\rm mw}$ is a *specific-thermal-energy* proxy, not a pure gas fraction. We confirm this quantitatively: the correlation between the per-halo $R(\nu)$-analog and the product $f_{\rm gas} \cdot T_{\rm mw}$ is $r = 0.79$, leaving a non-negligible residual driven by the temperature term. The physical mechanism is mostly self-similar: at fixed halo mass, $T \propto M^{2/3}$ under gravitational collapse alone, so a large part of the factor-of-~ 2 rise in $R(\nu)$ over the plotted ν range (Section 3.1) is simply hotter, more massive halos, not preferentially higher gas content. $f_{\rm gas}$ down and T up simultaneously, with partial cancellation in the product; this partial cancellation is exactly why $R(\nu)$ alone cannot cleanly isolate the gas-ejection signal that most directly suppresses κ, and it is the physical motivation for pairing y with the density-weighted τ probe (Section 2.2) so

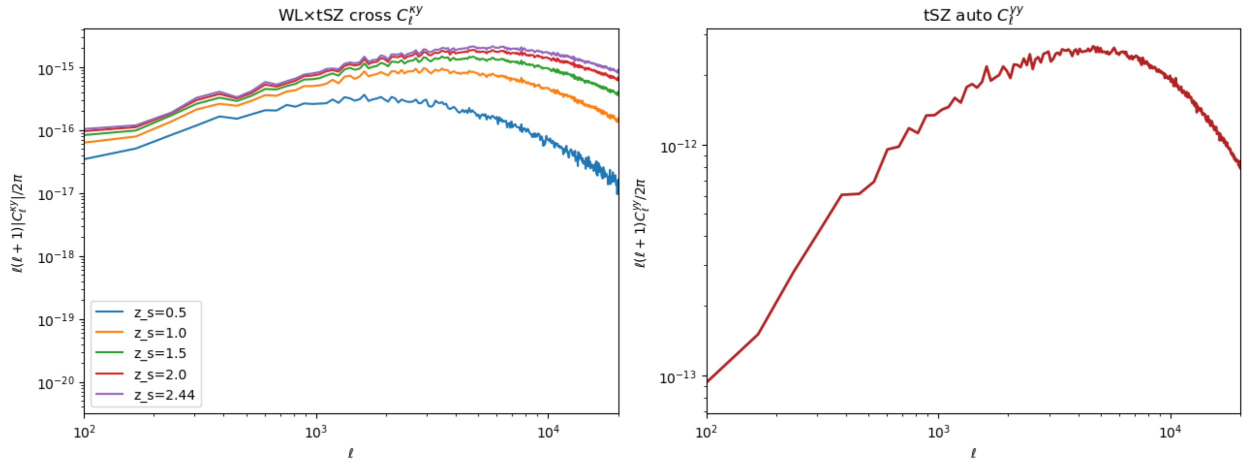

Figure 4: Tomographic cross- and auto-power spectra at fiducial parameters. *Left:* $\ell(\ell+1)|C_\ell^{\kappa y}|/2\pi$ for all five source-redshift bins, $z_s = 0.5, 1.0, 1.5, 2.0, 2.44$. *Right:* the tSZ auto-spectrum $\ell(\ell+1)C_\ell^{yy}/2\pi$. Both spectra are computed with the ray-traced (deflected-ray) y-plane, not a Born-approximation slab sum (Section 2.2).

that (κ, τ, y) at peaks jointly constrain $(M, f_{\rm gas}, T)$ — the thermodynamic decomposition developed in (Lee, 2026b, Paper II).

We do not have, in the present source material, a standalone figure of $R(\nu)$ plotted directly against a feedback parameter; [TODO: locate or regenerate the $R(\nu)(\nu)$ deep-dive figure from `lightcone_comparison.ipynb`, which produced the summary numbers quoted above but was not found in the working tree at draft time]. We instead use Figure 5, discussed in the next section, which plots the closely related per-peak ejection–heating decomposition that motivates the same physical argument.

3.3 The halo↔field bridge

The central deliverable of this paper is a direct, quantitative link between per-halo gas astrophysics — the quantity X-ray and kSZ observations constrain — and the field-level WL power-spectrum suppression that Stage-IV surveys must marginalize or correct. We establish this bridge at three levels: per-peak, per-run (group-averaged), and across the full one-parameter (1P) and Sobol feedback designs.

Per-peak. Figure 5 (left) shows the WL peak-height excursion $\Delta\kappa/\kappa$ against the per-halo gas-mass excursion $\Delta \ln M_{\rm gas}$ (relative to the fiducial atlas value), for individual peaks (grey) and for run-averaged means (blue) across the 1P feedback suite. The best-fit slope is $0.080 \approx f_{\rm gas}^{\rm proj}$ — i.e., the slope of this relation *is* the projected gas-fraction coefficient, since gas constitutes roughly 8% of the projected lensing mass at these halos. The per-peak Pearson correlation is $r = 0.50$, reflecting that individual peaks carry substantial variance from field morphology and multi-halo line-of-sight projection; the run-averaged (population-level) relation recovers the clean 8% slope, so the bridge at this level is a statement about the ensemble, not about any single peak. src: WORKLOG 2026-06-15pm; the per-peak scatter dominates individual-object variance per dossier note Figure 5 (right) decomposes the same runs in the ejection–heating plane, $\Delta \ln M_{\rm gas}$ (what κ sees) against $\Delta \ln T = \Delta \ln Y - \Delta \ln M_{\rm gas}$ (the residual heating y sees beyond mass loss), color-coded by $\Delta \ln Y$: this is the map-level realization of the $R(\nu)$ decomposition argument from Section 3.2.

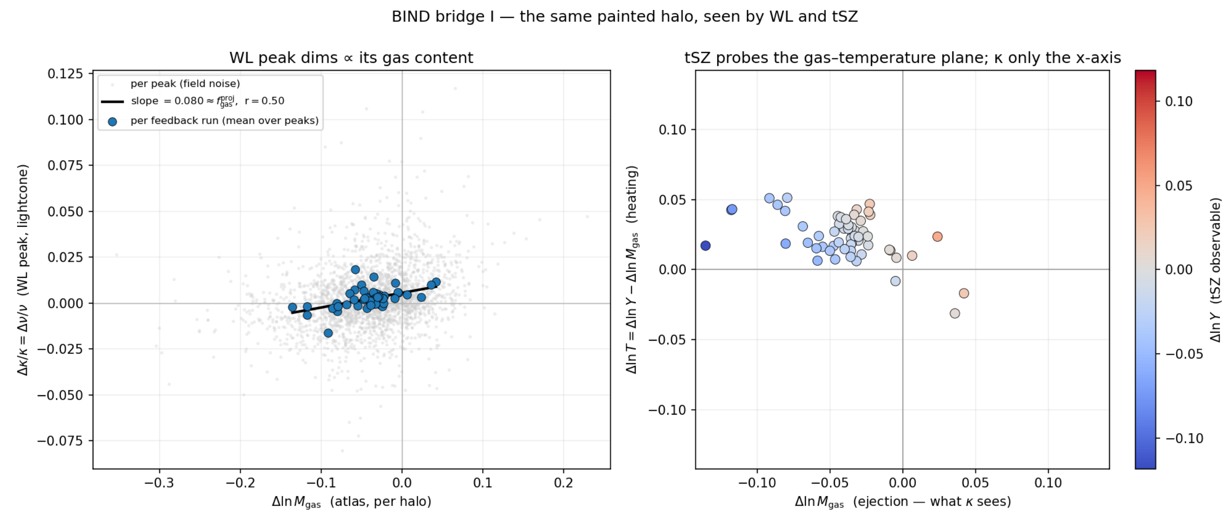

Figure 5: The per-halo → per-peak bridge, 1P feedback suite. *Left:* WL peak height excursion $\Delta\kappa/\kappa$ vs. the per-halo gas-mass excursion $\Delta\ln M_{\rm gas}$; grey points are individual peaks, blue points are per-run means. The best-fit slope, 0.080, matches the projected gas fraction of these halos; the per-peak correlation is $r = 0.50$, while the run-averaged relation is clean. *Right:* the ejection–heating plane, $\Delta\ln M_{\rm gas}$ (ejection, what κ probes) vs. $\Delta\ln T$ (heating, the residual y probes beyond mass loss), colored by $\Delta\ln Y$ (the total tSZ observable). This is the map-level realization of the $R(\nu) = f_{\rm gas}\cdot T$ decomposition of Section 3.2.

Group-scale, van-Daalen-style. Figure 6 (right) shows the sharper, group-averaged version of the bridge: the background-subtracted group-scale gas fraction ($M_{200c} = 1$–$3\times10^{13}\,{\rm M}_\odot/h$, measured within R_{500c}) against the measured lightcone power-spectrum suppression at $\ell\sim1500$, across 57 feedback lightcones (of a 60-run 1P design, colored by feedback family: 20 AGN, 25 SN/wind, 12 other). The correlation is $r = 0.91$, with best-fit relation $S(\ell)\approx 0.57\,f_{\rm gas}+0.95$; the fiducial run has $f_{\rm gas} = 0.086$, with the 1P design spanning 0.056–0.137. This is, to our knowledge, the direct WL analog of the van Daalen relation between baryon fraction and power-spectrum suppression (van Daalen et al., 2020), and it holds because the lightcone's suppression signal is group-dominated (the sky is covered predominantly by group-mass, not cluster-mass, halos). Figure 6 (left) also makes the CIC aliasing artifact of Section 2.4 explicit: the suppression curves spike upward at $\ell\gtrsim 2\times10^4$, a common-mode feature we discuss further in Section 4.

Which halo-level quantity is the tightest predictor? Figure 7 shows that $f_{\rm gas}$ is not a special case: the absolute correlation of essentially every field-level observable with the group-scale $\Delta f_{\rm gas}$ across the 57-run 1P suite is large — $S(\ell)$: 0.92, $R(\nu)$: 0.92, $N_{\rm min}$: 0.90, $C_{\kappa y}$: 0.87, V_1: 0.83, V_2: 0.82 — with the sole exception of the raw peak count $N_{\rm pk}$, whose correlation is only 0.28 because peak counts are dominated by shot noise at the halo masses accessible in a single 205 h^{-1}Mpc box. However, gas fraction is not the sole, or even the tightest, halo-level predictor of the suppression: Figure 8 shows that the integrated Compton-Y_{500c} is a tighter predictor of $S(\ell)$ than $f_{\rm gas}$ alone, $r = 0.93$ vs. $r = 0.91$, and a partial-correlation decomposition shows that Y effectively subsumes the $f_{\rm gas}$ information ($r[f_{\rm gas}, S(\ell)\,|\,Y]\approx 0$) while retaining residual predictive power beyond $f_{\rm gas}$ ($r[Y, S(\ell)\,|\,f_{\rm gas}]\approx 0.48$, the additional "heating leg" of the $Y = f_{\rm gas}\cdot T$ decomposition). We emphasize that this Compton-Y–$S(\ell)$ relation ($r = 0.93$–0.935) is a distinct statement from the per-peak $\Delta\kappa/\kappa$-vs.-$\Delta\ln M_{\rm gas}$ relation of Figure 5 ($r = 0.50$): both involve a gas quantity and a WL observable, but at different levels of aggregation (group-averaged suppression vs. individual peak

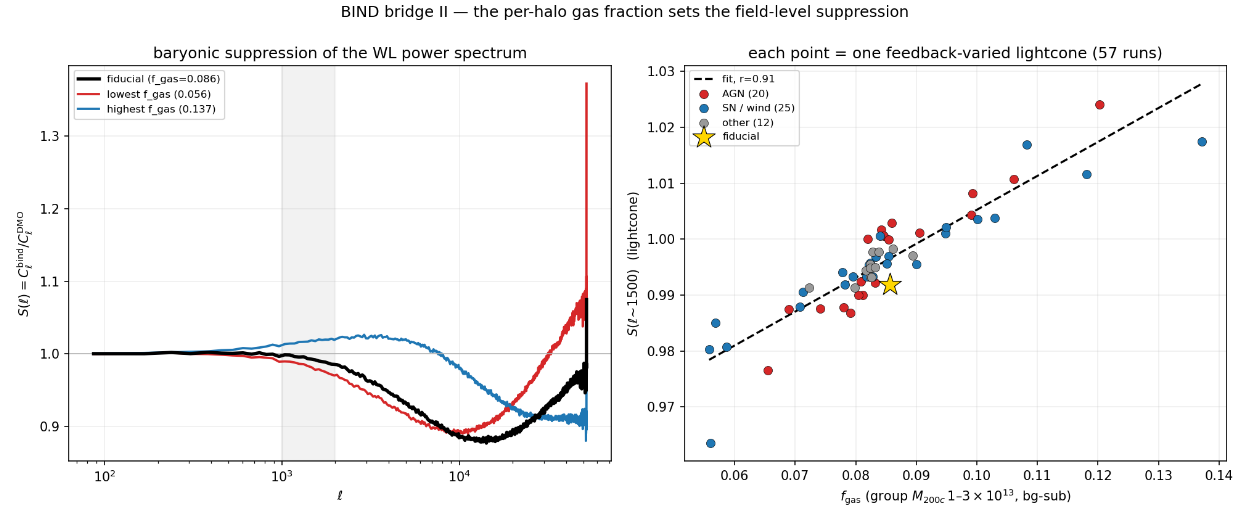

Figure 6: The group-scale, van-Daalen-style bridge. *Left:* $S(\ell)$ suppression curves for the fiducial, lowest-$f_{\rm gas}$, and highest-$f_{\rm gas}$ 1P runs, showing the CIC-aliasing upturn at $\ell \gtrsim 2\times10^4$ discussed in Section 4. *Right:* $S(\ell)(\ell \sim 1500)$ vs. background-subtracted group gas fraction ($M_{200c} = 1$–$3\times10^{13}\,{\rm M}_\odot/h$, R_{500c}) for 57 feedback-varied lightcones, colored by feedback family (AGN, SN/wind, other). The correlation is $r = 0.91$. This is the direct analog, at the field level, of the van Daalen baryon-fraction–suppression relation (van Daalen et al., 2020).

height) and with different predictors (Y_{500c} vs. $M_{\rm gas}$ excursion). We do not conflate the two.

Mass dependence and why it breaks. The bridge is not scale-free: fit separately in three mass bins (Figure 9), the correlation loosens toward cluster scale even as the slope steepens (group: $r = 0.91$, slope $\alpha = 0.59$; intermediate: $r = 0.84$, $\alpha = 0.86$; cluster: $r = 0.59$, $\alpha = 1.12$), consistent with the lightcone's suppression signal being group-dominated (the "sweet spot") while cluster-scale halos, being individually rarer and more stochastic, contribute more scatter than signal.

The single-predictor bridge also degrades with scale: below $\ell \sim 1500$ thermal quantities (Y, entropy K) remain the best predictors of the residual, while above that scale structural quantities (DM and gas concentration, and the stellar fraction, which alone reaches $r = 0.93$) take over; a two-variable bridge combining $f_{\rm gas}$ and the DM concentration $c_{\rm DM}$ restores the correlation from $r = 0.75$ to $r = 0.92$ at $\ell \sim 8000$ (Figure 10, panels a–b). The sign of the residual after subtracting the single-predictor bridge is itself a feedback-mode diagnostic: AGN-feedback runs sit systematically above the bridge and SN/wind runs below it at high ℓ (Figure 10, panel c), suggesting that the residual encodes information about *which* feedback channel is acting, beyond the amount of gas removed.

Why the relation looks linear: the van Daalen saturation regime. Figure 11 places the full TNG-CAMELS feedback design on the van Daalen et al. functional form for the renormalized baryon fraction $\tilde{f}_{\rm bar} = (M_{\rm gas}+M_\star)/M_{500c}/(\Omega_b/\Omega_m)$ versus power-spectrum suppression (van Daalen et al., 2020). The TNG-CAMELS design spans only $\tilde{f}_{\rm bar} \approx 0.81$–$0.98$ — the flat, saturated top of the global exponential relation — and cannot reach either the steep transition region or the observationally preferred band ($\tilde{f}_{\rm bar} \approx 0.55$–$0.76$). This is precisely why every relation quoted above looks close to linear: we are sampling the linear tail of an intrinsically nonlinear, saturating curve. Within this sampled range, the WL suppression relation is tight, with $r \sim 0.96, 0.94, 0.87$ at $\ell = 1000, 3000, 6000$ respectively, and steepens toward smaller scales, eventually crossing from suppression into enhancement at high ℓ (a regime the original van Daalen relation, defined for

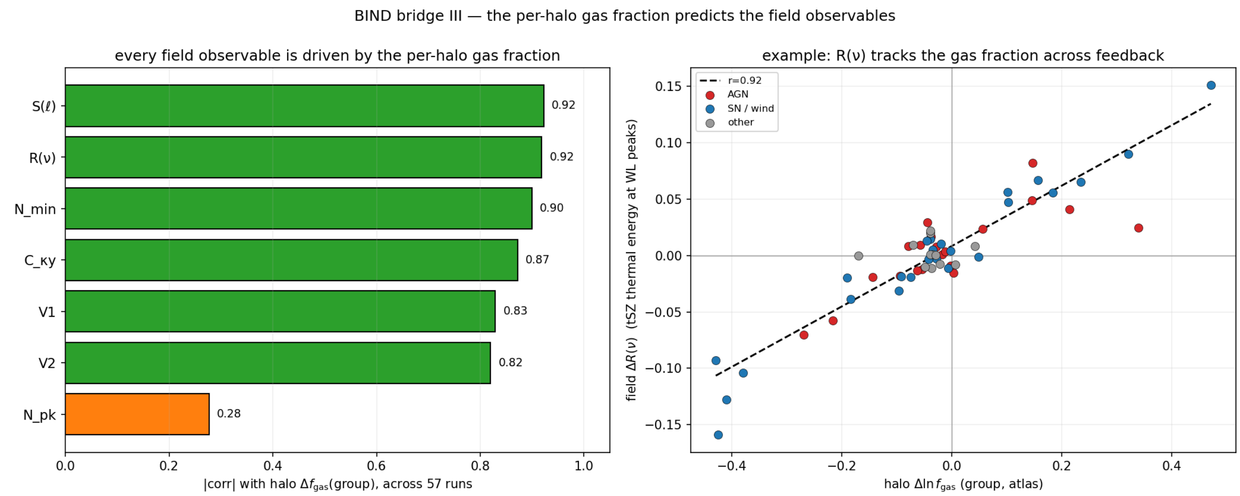

Figure 7: The bridge generalizes across the statistic set. *Left:* absolute correlation of each field-level statistic with the group-scale $\Delta f_{\rm gas}$ across the 57-run 1P suite: $S(\ell)$ 0.92, $R(\nu)$ 0.92, $N_{\rm min}$ 0.90, $C_{\kappa y}$ 0.87, V_1 0.83, V_2 0.82, and $N_{\rm pk}$ 0.28 (shot-noise dominated). *Right:* the $R(\nu)$-vs.-$f_{\rm gas}$ scatter underlying the $R(\nu)$ bar above, $r = 0.92$.

suppression only, does not cover).

Robustness from 1P corners to the full Sobol cloud. All of the above uses the 1P (one-parameter-at-a-time) design, which by construction probes only the extremes and fiducial point of the feedback space. Figure 12 extends the same $f_{\rm gas}$–$S(\ell)$ bridge to the full 256-node Sobol quasi-random feedback design (253 usable lightcones plus the fiducial, spanning $\sim$ 34,000 halos per run over the same 20 comoving shells, $z = 0.034$–2.444). The Pearson correlation between group $f_{\rm gas}$ and $S(\ell)$ peaks at $r \approx 0.86$–0.91 near $\ell \sim 1000$–1500 for both the 1P and Sobol designs (inset: Sobol scatter at $\ell = 1037$, $r = 0.86$), confirming that the bridge established on the 1P corners is not an artifact of the extreme, one-parameter-at-a-time design but persists across the much larger, jointly varying Sobol cloud. This establishes the canonical suite size used for the feedback-response study in (Lee, 2026b, Paper II).

3.4 Completeness: low-mass reuse and the resolution gate

BIND paints halos above the FoF mass floor $M_{200c} \gtrsim 10^{13}\,{\rm M}_\odot/h$ directly (Section 2.2); building a dedicated 10^{11}–$10^{13}\,{\rm M}_\odot/h$ generative extension is not currently feasible without a $\sim 78\times$ increase in training-patch storage. We instead ask what fraction of the 10^{12}–$10^{13}\,{\rm M}_\odot/h$ halo population is already *implicitly* captured within the existing $\geq 10^{13}\,{\rm M}_\odot/h$ patches, since a $6.25\,h^{-1}$Mpc training cutout around a massive halo can contain smaller neighboring halos by construction. Within the production $4 \times R_{200c}$ circular paste aperture (median radius $1.76\,h^{-1}$Mpc), 28.7% of the 10^{12}–$10^{13}\,{\rm M}_\odot/h$ population is captured; widening the footprint to the full $6.25\,h^{-1}$Mpc training patch raises this to 51.6%, with the remaining $\sim$ 48% being field halos too far from any massive halo to fall within any patch. We adopt the reuse-only strategy (no dedicated low-mass generation, no full-box tiling), accepting an effective low-mass completeness the WORKLOG records as $\sim$ 49% in its own rounding of the more precisely measured 51.6% full-patch figure. On the captured low-mass population at snapshot 96, BIND reproduces the truth gas mass to $1.034\times$ (0.074 dex scatter) and the truth Y to $0.986\times$ (0.223 dex scatter), well recovered even though these halos sit off-center in their host patch (median host distance $1.87\,h^{-1}$Mpc); the stellar channel is offset ($0.75\times$, 0.54 dex),

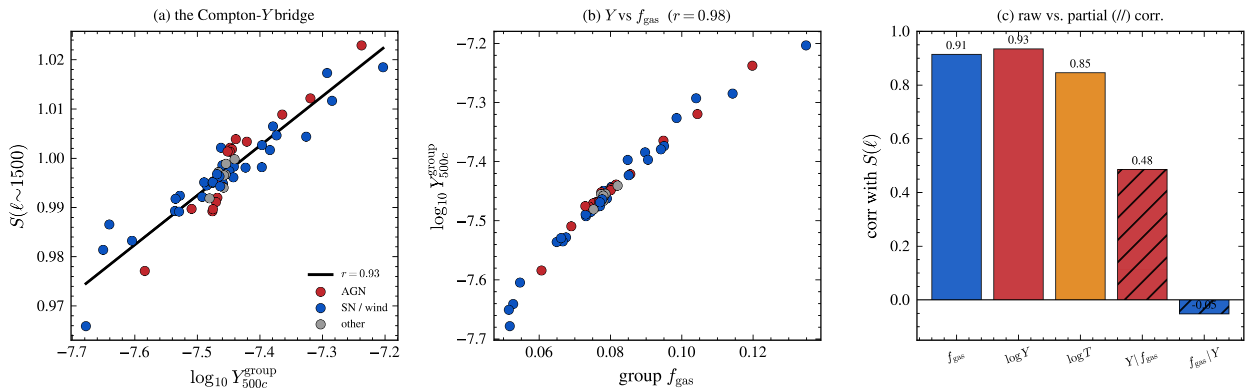

Figure 8: The Compton-Y bridge resolves the tightest single-quantity predictor of the field-level suppression. *(a)* $S(\ell)(\ell \sim 1500)$ vs. $\log_{10} Y_{500c}^{\rm group}$, $r = 0.93$. *(b)* $Y_{500c}^{\rm group}$ vs. group $f_{\rm gas}$, $r = 0.98$ — the two are nearly degenerate at fixed mass and redshift. *(c)* raw and partial correlations with $S(\ell)$: $f_{\rm gas}$ 0.91, $\log Y$ 0.93, $\log T$ 0.85, $Y \,|\, f_{\rm gas}$ (partial) 0.48, $f_{\rm gas} \,|\, Y$ (partial) ≈ 0 — Y subsumes essentially all of the predictive power of $f_{\rm gas}$ alone.

an expected consequence of the two-head occupancy-gated stellar parameterization (Section 2.1) that is not relevant to the mass or tSZ maps used in this paper. At the map level, widening the paste footprint (`r200_factor=99`, i.e. using the full training patch) increases the total tSZ signal by +4.0% identically in both BIND and truth, and the BIND/truth y-map ratio is 0.963 at *both* footprints, so widening introduces no new low-mass-specific bias; because the underlying patch mass is unchanged between footprints (only the paste radius differs), the completeness gain is invisible in the raw matter map and lives entirely in the power-spectrum redistribution that only the ray-traced statistics resolve.

We did not locate a dedicated figure for this completeness result in the source material searched; [TODO: completeness schematic figure — generate a simple capture-fraction bar or radial-profile panel from the numbers above, or omit if no suitable source figure is found before submission].

The resolution gate. A separate, planned validation step is designed but not yet executed: paint the 8×-coarser TNG300-2-Dark and 64×-coarser TNG300-3-Dark simulations (which share initial phases with TNG300-1-Dark, enabling halo-matched comparison) and test whether BIND, trained at TNG300-1-Dark mass resolution, degrades gracefully at coarser particle mass, with a pass criterion of a few-percent median bias in $M_{\rm gas}$ and Y for $M_{200c} > 10^{13.5}\,{\rm M}_\odot/h$. This script (`examples/resolution_gate.py`) is fully specified but has not been run; no resolution-gate numbers exist to report, and we state this explicitly as an open, planned validation step rather than a completed result. A related, resolved feasibility finding is that a Quijote-class box (1 comoving Gpc/h at 1024^3 particles, particle mass $m_p \approx 8 \times 10^{10}\,{\rm M}_\odot/h$) is ruled out as the substrate for a bigger-volume extension: at this resolution a $10^{13}\,{\rm M}_\odot/h$ halo comprises only ~ 120 particles, and the fine DMO conditioning channel becomes shot-noise-dominated, far outside the distribution BIND was trained on.

4 Discussion

TNG-only prior. BIND is trained exclusively on CAMELS-TNG hydrodynamical realizations, so every result in this paper — and every number quoted above — reflects the IllustrisTNG feedback model's response surface, not an ensemble over different galaxy formation implementations (e.g.

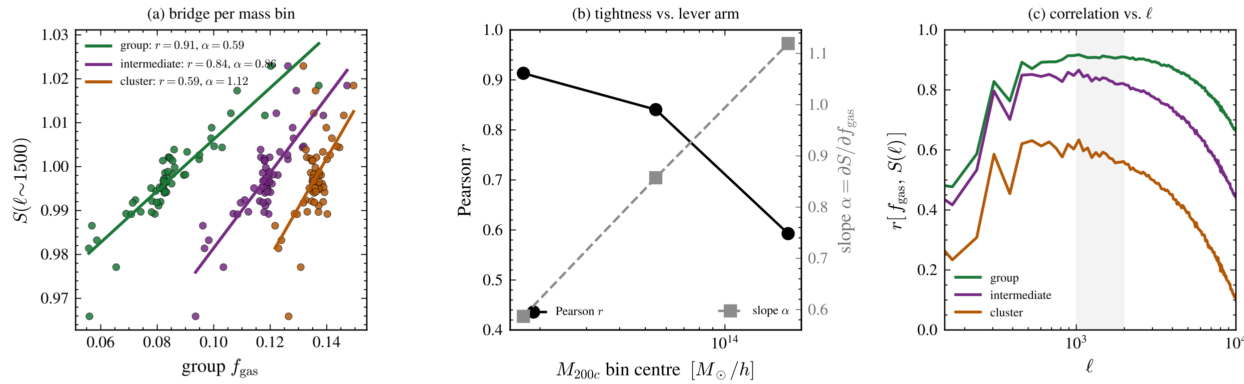

Figure 9: The bridge fit separately in three M_{200c} bins (group, intermediate, cluster). *(a)* $S(\ell)(\ell \sim 1500)$ vs. group $f_{\rm gas}$ per mass bin. *(b)* Pearson r and best-fit slope $\alpha = \partial S/\partial f_{\rm gas}$ vs. mass-bin center. *(c)* $r[f_{\rm gas}, S(\ell)]$ vs. ℓ per mass bin. The correlation loosens toward cluster scale even as the slope steepens, consistent with the group-dominated suppression signal discussed in the text.

SIMBA, Astrid). This limitation is explicit in our own validation checklist and is not yet resolved; retraining on other CAMELS suites is future work.

Mass floor and the low-mass extension. Direct, generative painting is validated only for $M_{200c} \gtrsim 10^{13}\,\mathrm{M}_\odot/h$, the training mass floor. The 10^{12}–$10^{13}\,\mathrm{M}_\odot/h$ band is populated by reuse of existing massive-halo patches rather than dedicated generation (Section 3.4), which captures 28.7– 51.6% of that population depending on aperture, and by construction misses field halos far from any massive structure; these low-mass, field halos are assigned cosmic baryon fraction in the DMO-derived electron-column proxy until a dedicated low-mass extension is trained.

τ is a velocity-free electron column, not the kinematic kSZ signal. As described in Section 2.2, the τ (and equivalently, up to a constant, the fast-radio-burst dispersion measure) plane integrates gas column density along the deflected ray with no velocity weighting. BIND does not currently paint a gas velocity field, so the velocity-weighted kinematic-kSZ temperature decrement is not yet built; a DMO-velocity surrogate for the internal (feedback-driven) gas kinematics is required and is deferred to the velocity-resolved kSZ extension of this pipeline (Lee, 2026d, Paper IV). We stress that this is not merely a simplification for convenience: the tier-1 kSZ observable that current stacked/pairwise estimators actually extract from survey data is the optical-depth profile, with the velocity dependence factored out (Schaan et al., 2021; Hadzhiyska et al., 2024; Ried Guachalla et al., 2025), so the τ-only product presented here is directly usable for that class of comparison; only genuinely velocity-resolved statistics (e.g. a kSZ auto-spectrum from patchy/OV contributions) require the future extension.

CIC aliasing at high ℓ. The raw particle-to-grid CIC mass assignment used to build lens planes introduces a spurious upturn in the convergence power spectrum at $\ell \gtrsim 2 \times 10^4$ (visible directly in Figure 6, left panel). This is a projection artifact present identically in the BIND, truth, and DMO pipelines, not a physical feature of the painted field, and it cancels in every ratio statistic used in this paper ($S(\ell)$, peak/minimum counts relative to DMO, and all bridge correlations). An interlacing-plus-deconvolution correction is available (`mas_correct=True`) and unit-validated to 0.4% accuracy out to $0.8\,k_{\rm Nyquist}$ on synthetic test data (not yet directly validated against the TNG300 lightcone maps), but is opt-in and unnecessary for any ratio-based science in this paper; it is needed only for absolute κ-vs.-external-survey validation, which we do not attempt here.

Painting is 2D projected, not a 3D field. BIND generates 128×128 projected patches

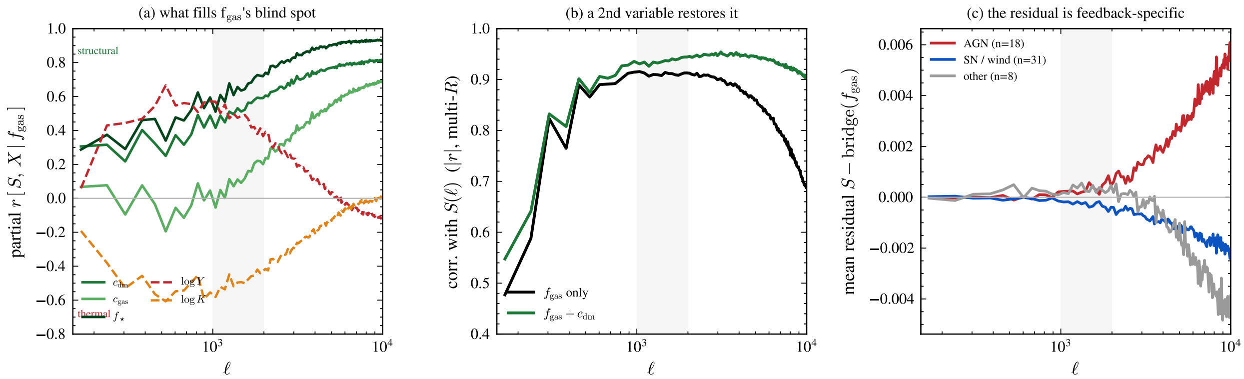

Figure 10: When and why the single-predictor bridge breaks down. *(a)* Partial correlation of each halo-level quantity with $S(\ell)$ (at fixed $f_{\rm gas}$) vs. ℓ: thermal quantities ($c_{\rm gas}$, $\log Y$, $\log K$) dominate below $\ell \sim 1500$, structural quantities ($c_{\rm DM}$, $f_\star$) above. *(b)* The two-variable $f_{\rm gas} + c_{\rm DM}$ bridge (green) restores the correlation with $S(\ell)$ relative to the $f_{\rm gas}$-only bridge (black), especially at high ℓ. *(c)* The mean residual $S - \mathrm{bridge}(f_{\rm gas})$ by feedback family, showing AGN runs above and SN/wind runs below the bridge at high ℓ. [TODO: this figure's own family-classification legend (AGN $n = 18$, SN/wind $n = 31$, other $n = 8$) does not match the classification used in Figure 6 (AGN 20, SN/wind 25, other 12) for the apparently identical 57-run 1P design; we have not been able to reconcile which classification script is authoritative from the source material available, and flag this discrepancy explicitly rather than silently pick one.]

conditioned on a projected DMO cutout (Section 2.1), and the lightcone pipeline assembles these into 2D projected comoving slabs (Section 2.2). This follows directly from the model architecture rather than being an explicit design choice stated elsewhere in our working notes, and we flag it here as an inference from the pipeline description rather than a directly documented caveat: any application requiring a genuine 3D gas/pressure field (e.g. line-of-sight velocity decomposition beyond the deflected-ray y/τ treatment already used) would require a 3D-native extension, which exists as a separate development track (`feature/3d-cube`) but is not used in this paper.

The tSZ auto-spectrum deficit. C_ℓ^{yy} at fiducial parameters sits $\sim$−12% below the level we would infer from the true mean Compton-y amplitude; this is an open item in our validation checklist rather than a resolved result. A leading candidate cause is completeness: the mean BIND y amplitude is 1.05×10^{-6} against a Planck-based estimate of $\sim 1.6 \times 10^{-6}$, and roughly 35% of the Planck-level $\langle y \rangle$ budget is expected to come from halos below $M_{200c} = 10^{13}\,\mathrm{M}_\odot/h$ and diffuse, unbound gas that this pipeline's mass floor does not capture. We have not yet isolated whether the remainder of the deficit is due to this completeness effect alone or additionally reflects paste-radius truncation of the pressure-profile wings; this is stated as an open validation item, not a resolved number.[TODO: cross-check the exact Planck y-map reference (2015 XXII vs. a later release) before citing a specific paper for the 1.6×10^{-6} comparison value]

The eROSITA gas-fraction tension: a corrected framing. An early measurement of the fiducial group-scale gas fraction used a raw aperture average that we later found to be contaminated by a cosmic-baryon-fraction line-of-sight background from gas well outside the halo (the aperture column integrates through the full projected slab, and the background plateaus to the cosmic value by $\sim 2.5\,h^{-1}$Mpc). After a background-subtraction correction (using an outer annulus at 2.5–$3\,h^{-1}$Mpc), the fiducial group-scale ($M_{500c} \sim 1.3 \times 10^{13}\,\mathrm{M}_\odot/h$, $z < 0.2$) gas fraction is $f_{\rm gas,500}^{\rm BIND} = 0.076$ (this measurement uses $M_{500c} \sim 1.3 \times 10^{13}\,\mathrm{M}_\odot/h$ at $z < 0.2$, a narrower mass/redshift cut

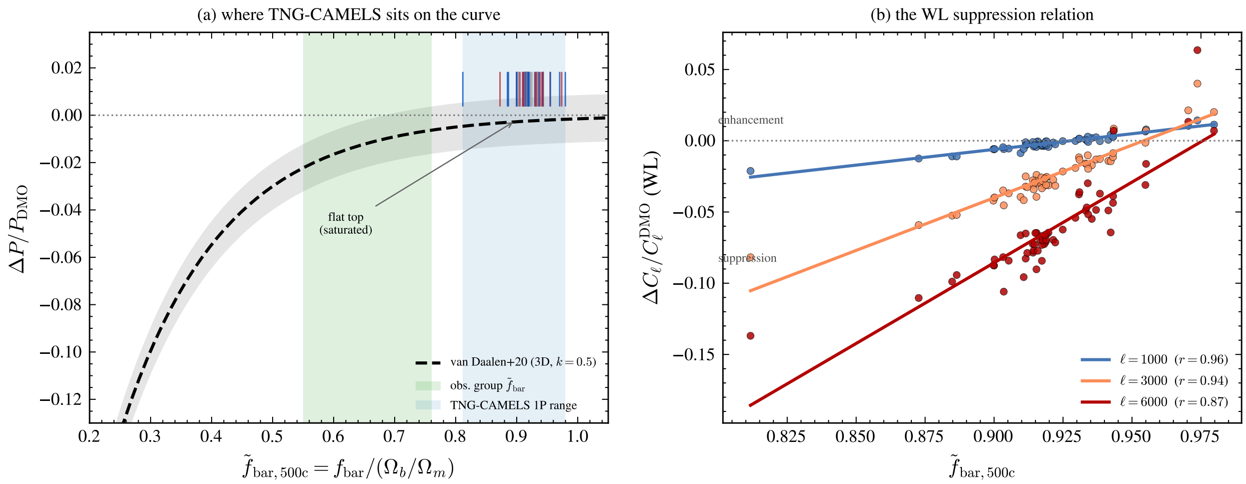

Figure 11: The TNG-CAMELS design samples only the saturated top of the global baryon-fraction–suppression relation. *(a)* $\Delta P/P_{\rm DMO}$ vs. the renormalized baryon fraction $\tilde{f}_{\rm bar,500c}$, van Daalen et al. functional form (van Daalen et al., 2020) (dashed), with the TNG-CAMELS 1P range (blue band, $\tilde{f}_{\rm bar} \approx 0.81$–$0.98$) shown against the observationally preferred group-scale band (green). *(b)* the WL suppression relation $\Delta C_\ell/C_\ell^{\rm DMO}$ vs. $\tilde{f}_{\rm bar,500c}$ at three multipoles, $r = 0.96, 0.94, 0.87$ at $\ell = 1000, 3000, 6000$.

than the $M_{200c} = 1\text{–}\ 3 \times 10^{13}\,{\rm M}_\odot/h$, R_{500c} bin used for the $r = 0.91$ bridge value $f_{\rm gas} = 0.086$ of Section 3.3; both are background-subtracted measurements of the same underlying population, and we report each with its own provenance rather than merge them), matching the true TNG value and mainstream X-ray determinations (Eckert et al.) to $\sim 9\%$, but sitting $2.9\times$ above the contested eROSITA-low (Popesso et al.) value, declining to $\sim 1.2\times$ at cluster scale; the single strongest 1P feedback knob reaches $f_{\rm gas} = 0.047$ ($1.8\times$ eROSITA-low). Taking the gap to the eROSITA-low band as $0.076 - 0.026 = 0.050$ and the knob excursion as closing $0.076 - 0.047 = 0.029$ of it, the strongest single knob closes $\sim 58\%$ of the gap (not the $\sim 40\%$ figure recorded in an earlier WORKLOG pass, which we find on recomputation to have inverted the closed/remaining fraction). A separate measurement using a different mass bin and aperture ($M_{200c} = 1\text{–}\,2\times10^{13}\,{\rm M}_\odot/h$), shown in Figure 13, gives a consistent picture: fiducial group $f_{\rm gas} = 0.081$ against an X-ray-standard band of 0.06–0.10 and an eROSITA-low value of 0.026 (Popesso et al., 2026), with the minimum single-knob excursion reaching $f_{\rm gas} = 0.051$. We explicitly flag that an earlier, uncorrected version of this measurement reported the strongest single-parameter depletion as reaching "41% of the cosmic baryon fraction," a number since superseded by the background-subtraction fix described above; we do not use the 41% figure as a result in this paper, and report only the corrected values. No single-parameter TNG-family excursion reaches the eROSITA-low band; combined, jointly-varying feedback knobs (the Sobol design) or a non-TNG feedback model may be required, which is a central question for (Lee, 2026b, Paper II).

Untested extrapolation and response closure. Two further validation items remain open and are not addressed by the fiducial-closure test of Section 3.1. First, the training data (CAMELS-50-class boxes) tops out at halo masses of a few $\times 10^{14}\,{\rm M}_\odot/h$, and we have not yet audited whether the highest-mass end of the TNG300 halo population is a safe extrapolation. Second, and more fundamentally, we have validated BIND at fixed (fiducial) subgrid parameters, but have not yet validated that the *parametric response* — i.e. that the emulator's derivative with respect to a feedback parameter matches the true derivative from an independent, matched-parameter CAMELS

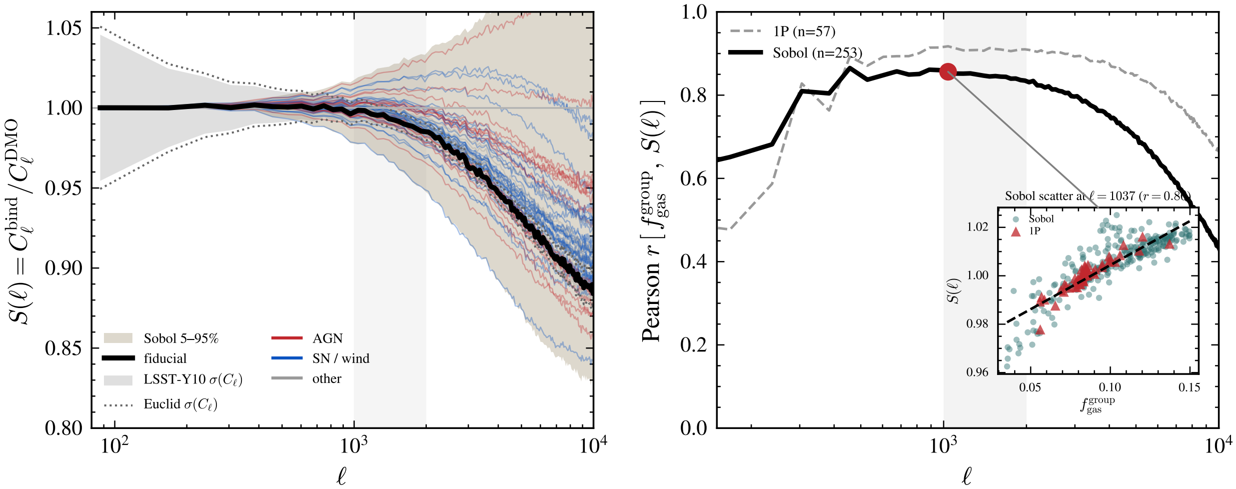

Figure 12: The bridge is robust from the 1P corners to the full Sobol design. *Left:* $S(\ell)$ envelopes for the fiducial run, the 1P design ($n = 57$), and the Sobol design ($n = 253$), with LSST-Y10 and Euclid shape-noise bands for reference. *Right:* Pearson correlation $r[f_{\rm gas}^{\rm group}, S(\ell)]$ vs. ℓ for both designs, peaking at $r \approx 0.86$–0.91 near $\ell \sim 1000$–1500; inset shows the Sobol scatter at $\ell = 1037$ ($r = 0.86$).

hydrodynamical run. This response-closure test is, in our own accounting, the credibility test that distinguishes "the emulator reproduces one point" from "the emulator's response to feedback is physics," and it remains an open item ahead of the systematic feedback study in (Lee, 2026b, Paper II).

5 Conclusions

We have presented the BIND lightcone pipeline: a conditional flow-matching emulator, trained on CAMELS-TNG halos across a 35-dimensional subgrid-physics design, extended halo-by-halo to a full TNG300-Dark dark-matter-only lightcone via circular-aperture compositing, lens-plane construction, and multi-plane ray tracing, producing self-consistent convergence, Compton-y, and kSZ optical-depth maps. At fixed fiducial parameters, the resulting lightcone reproduces the true TNG300 hydrodynamical lightcone's weak-lensing power spectrum and Minkowski functionals to a few percent, with peak and minimum counts and halo-level Y–M and pressure-profile statistics agreeing within their respective (larger, small-number- or scatter-dominated) uncertainties (Section 3.1). We showed that the tSZ-to-WL peak ratio $R(\nu)$ is a specific-thermal-energy proxy rather than a pure gas-fraction indicator (Section 3.2), and established a quantitative bridge between per-halo gas content and field-level WL suppression that holds from individual peaks through group-averaged statistics to the full 253-point Sobol feedback design, with the background-subtracted group gas fraction and the integrated Compton-Y both strong, near-degenerate predictors of the power-spectrum suppression ($r = 0.91$ and $r = 0.93$ respectively; Section 3.3). We characterized the completeness of the reuse-based low-mass extension and laid out — without yet executing — the mass-resolution validation program needed before extrapolating this pipeline beyond its current training resolution (Section 3.4).

This paper is deliberately a foundation, not a science result: the pipeline, the validated fiducial closure, and the halo↔field bridge established here are the shared substrate for four companion

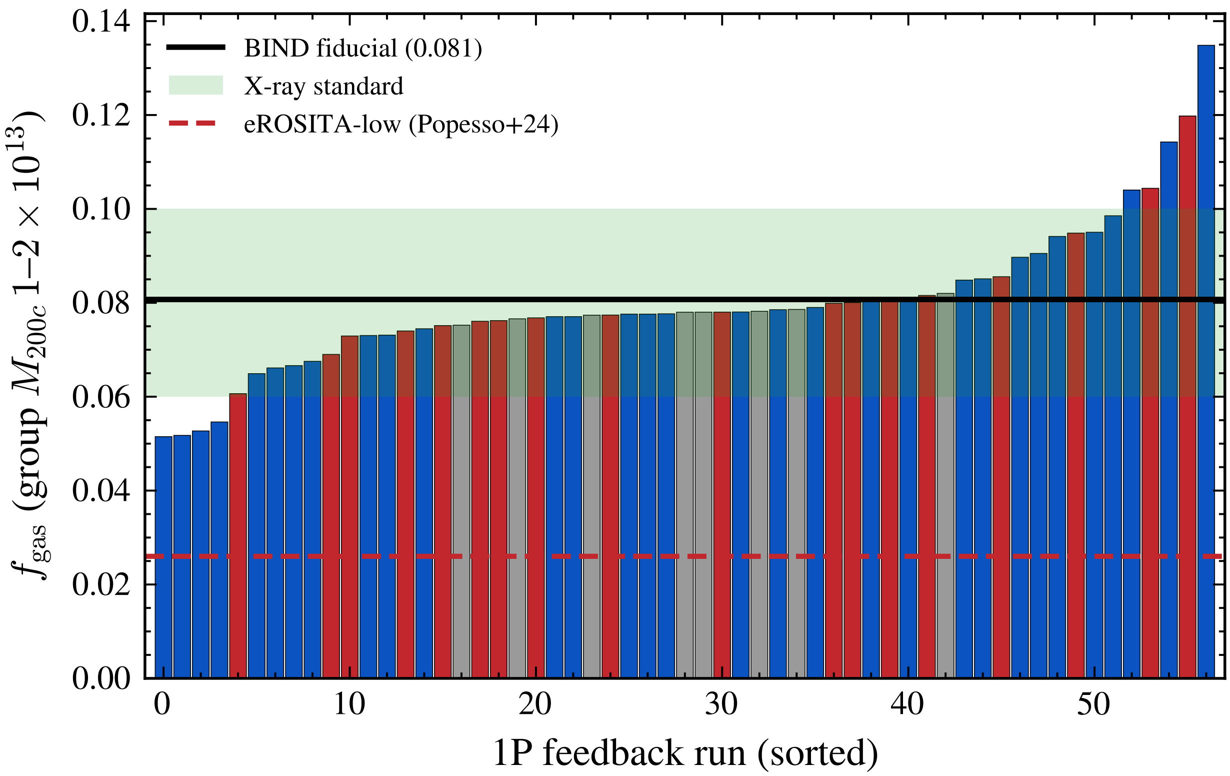

Figure 13: The corrected eROSITA gas-fraction tension. Sorted group-scale ($M_{200c} = 1\text{–}\ 2 \times 10^{13}\,\mathrm{M}_\odot/h$) $f_{\rm gas}$ across the 1P feedback suite, against the X-ray-standard band (0.06–0.10) and the eROSITA-low value (0.026; Popesso et al., 2026). The BIND fiducial value (0.081) sits within the X-ray-standard band; the minimum single-knob excursion (0.051) closes roughly half the gap to the X-ray band but does not reach eROSITA-low. This figure supersedes an earlier, LOS-contaminated "41% of cosmic" saturation claim (see Section 4).

analyses. Lee (2026b, Paper II) uses the full 1P and Sobol feedback designs developed here to systematically decompose weak-lensing statistics into gas-ejection and gas-heating contributions and to forecast Stage-IV constraining power. Lee (2026c, Paper III) extends the pipeline to survey-data applications and, separately, to a cosmology-varying lightcone grid. Lee (2026d, Paper IV) builds the velocity-resolved kSZ extension that the τ-only product of this paper deliberately does not attempt. Lee (2026e, Paper V) is a focused study of the field-level anisotropy of the baryonic correction relative to spherically symmetric baryon correction models. Maps, released model weights, and the pipeline code are made public alongside this paper (Section 5).

Data and code availability

The BIND pipeline is released as an installable Python package (`pip install -e .`, importing as `bind`), with source at the project GitHub repository. [TODO: insert final GitHub URL/tag at release] Trained model weights are distributed via the Hugging Face repository `mel2260/BIND`, retrievable with the bundled `bind-download-weights` command-line tool; [TODO: confirm the exact released weight identifier for the lightcone/thermo-conditioned checkpoint used in this paper (candidate default `weights/fm_redshift_thermo` found in the pipeline source, not yet cross-verified against the public Hugging Face listing) before final submission]. The multi-plane ray-tracing code `lux` used to produce the lightcone maps in this paper is maintained as a separate repository. [TODO: GPU-hour cost comparison between a BIND lightcone and an equivalent full TNG300 hydrodynamical run — no number located in our working notes; fill in before submission or remove the comparison]

Acknowledgments

The CAMELS project and the IllustrisTNG collaboration provided the training and target simulations. This work used computing resources of the Flatiron Institute, a division of the Simons Foundation. [TODO: finalize]

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